Bonded knots

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1 Detection of Diagram-Reducing Strand Flip Move

1.1 Terminology

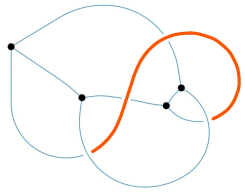
In the following, a *strand* refers to a simple path in the diagram whose endpoints are nodes (vertices or crossings) and whose interior contains no vertices. Crossings may occur along the strand.

A *diagram reduction* occurs when a diagram with n nodes can be transformed via Reidemeister moves into an equivalent diagram that contains strictly fewer than n nodes.

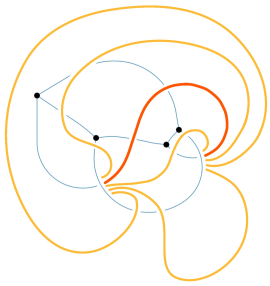
1.2 Idea

When a strand crosses exclusively over all other strands it encounters in the diagram (Fig. 1), it is possible to perform a *strand flip* - move the strand to an alternative position in the diagram in such a way that all of its current crossings disappear. This operation may introduce new crossings over other strands, depending on the chosen placement (Fig. 2a). Consequently, if the alternative position induces fewer crossings than the original, then the total number of crossings in a diagram decreases and the diagram is therefore reduced (Fig. 2b, 2c).

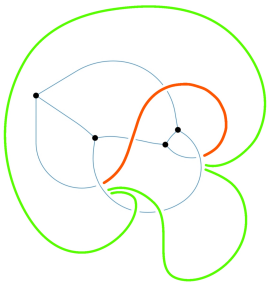
The idea above is described for strands that pass exclusively over other strands they encounter. By symmetry, the same reasoning holds for strands that pass exclusively under all encountered crossings.



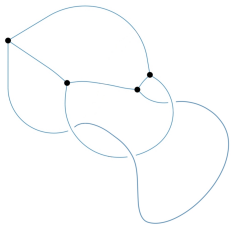
Rysunek 1: over-over path example



(a) Alternative path placement example



(b) better paths example



(c) Diagram after reduction

Rysunek 2: Caption

1.3 Detection Algorithm

For simplicity, the following algorithm description focuses solely on strands that pass exclusively over crossings and is analogous to the case of strands passing exclusively under crossings.

For every pair of nodes in a diagram (vertices and crossings):

1. If paths passing exclusively over crossings exist, find the one containing the largest number of crossings.

2. Compute all possible paths between the nodes and find the one that contains the smallest number of crossings.

3. If the number of crossings in the minimal-crossing path (2) is strictly smaller than in the maximal over-type path (1), a diagram reducing strand flip move is possible.

1.3.1 Finding an Over-Type Path

Finding an 'over'-type path between two nodes is performed by following each strand incident to the starting node and tracing it through the diagram. The process terminates either unsuccessfully, if a vertex or an 'under' crossing is encountered, or successfully if the target node is reached. In the latter case, the traced path is the desired 'over'-type path.

1.3.2 Finding a Minimal-Crossing Path

Computing an alternative placement for a strand, together with its cost, which represents the number of crossings that would be created by inserting a strand along this path, can be obtained by:

1. Following an arc until a crossing or a vertex is reached
2. At a vertex, the two remaining arcs must be considered. These correspond to continuing the traversal with either a left or a right turn.
3. At a crossing, three arcs may be followed: one to the left, one to the right and one passing straight through the crossing.

The side of the path along which the strand is inserted determines whether a crossing is created. Taking this into account, the following cost model is defined:

- Turns at vertices or crossings have cost 1 if the side changes and cost 0 otherwise.
- Passing straight through a crossing preserves the side and has cost 1.

The minimal-crossing path is then obtained by selecting the path with the minimal total cost according to this cost model.